T.S.A. of the hollow cylinder = C.S.A. + Area of two rings at the top and bottom.

$$= 2\pi(R + r)h \quad + 2\pi(R^2 - r^2)$$

T.S.A. of a hollow cylinder $= 2\pi(R + r)(R - r + h)$ sq. units

Example 7.4 If one litre of paint covers 10 m^2, how many litres of paint is required to paint the internal and external surface areas of a cylindrical tunnel whose thickness is 2 m, internal radius is 6 m and height is 25 m.

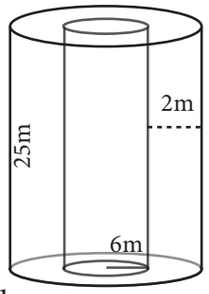


Fig. 7.8

Solution Given that, height $h = 25$ m; thickness $= 2$ m

internal radius $r = 6$ m

Now, external radius $R = 6 + 2 = 8$ m

C.S.A. of the cylindrical tunnel = C.S.A. of the hollow cylinder

C.S.A. of the hollow cylinder $= 2\pi(R + r)h$ sq. units

$$= 2 \times \frac{22}{7}(8 + 6) \times 25$$

Hence, C.S.A. of the cylindrical tunnel = 2200 m^2

Area covered by one litre of paint = 10 m^2

Number of litres required to paint the tunnel $= \frac{2200}{10} = 220$

Therefore, 220 litres of paint is needed to paint the tunnel.

Progress Check

1. Right circular cylinder is a solid obtained by revolving _____ about _____.
2. In a right circular cylinder the axis is _____to the diameter.
3. The difference between the C.S.A. and T.S.A. of a right circular cylinder is _____.
4. The C.S.A. of a right circular cylinder of equal radius and height is _____ the area of its base.

7.2.3 Right Circular Cone

Observe the given figures in Fig.7.9 and identify which solid shape they represent?

These objects resemble the shape of a cone.



Fig. 7.9

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